Swimming in the Spree

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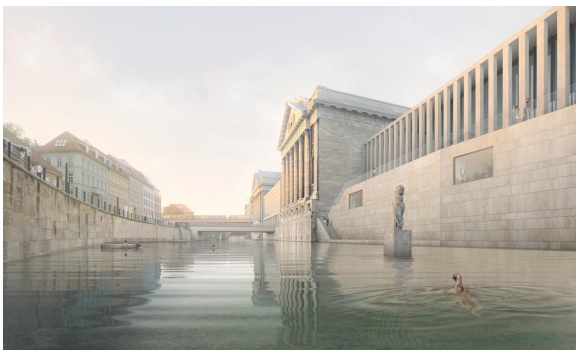
Background

Like every unique water to each city, the Spree witnessed a lot: the glacier that carved out the Berlin-Warsaw valley, the Prussian merchant ships, rubbles from air raids during WWII, people swimming to get across the Berlin wall, deserted Eastern factories and lignite mines...

Strolling by the Spree on a September afternoon

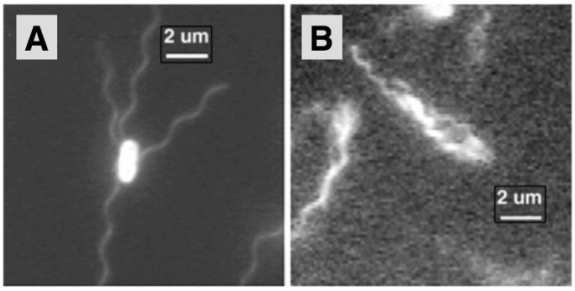
Before we can bathe in the Spree (like pictured), we can study the bacteria that currently swim in it, so that we can catch them all with filters and clean the water.



A swimmable Spree, imagined by Flussbad Berlin

Problem

What is the physical mechanism by which bacteria such as E Coli propel itself through the water?



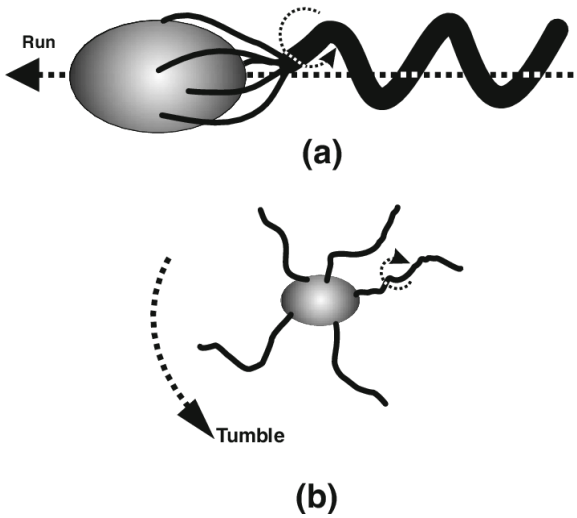
E.Coli motion (Kumar, 2009).

Methods

I actually came across a friendly physicist who studies bacteria swarms and other active fluids, but after attending one of their seminars I could not understand the language and tools they use, involving lots of differential equations and phase diagrams. I decided to figure out the simpler explanation by reasoning on my own with classical mechanics principles and then reading the common formulation.

Intuition

First I will try to guess the physical process governing a bacteria swimming.



E.Coli motion (Kumar, 2009).

A very tiny swimmer in a very sticky fluid moves by actively stroking instead of gliding. There is no inertia that lets it stay in motion.It wants to find higher concentrations of nutrients, closer to the surface, closer to light and warmth and optimal pH.

There are two main sources of motion:

1. Self propulsion. E Coli has a body and a bunch of flagellum. It rotates its flagella and its body in opposite directions to move forward. As a reaction, the liquid drags the cell in the opposite direction.
2. Brownian motion. Bacteria are very small (typically micrometer in size) so it is colliding into a lot of molecules and randomly jiggling.

We can look at the symmetry and continuity in the system — bacteria — in the medium.

Conservation of momentum: when we do not consider heat dissipation, there is no impulse that exchange momentum between the bacteria and the surrounding. When I swim, by Newton's second law - conservation of momentum -, when I push the water back, the water pushes me forward with the same momentum. But the bacteria is much smaller so when it swims in water there is a stronger resistance that is sticking it to where it is rather than letting it continue its motion.

$$\sum M = 0$$
$$M = mv$$

It has to produce some sort of net torque that lets the liquid counteract it and cause net displacement. The constant difference etween the angular momentum of the flagellum moves the bacteria forward.

$$M2 - M1 = const$$

$$\tau = F \times r$$

The propelling force—thrust— comes from the rotation of the flagella around the axis and gets counteracted by the resistive force—drag. Drag is not a conservative force but here we consider the whole system with surrounding contained.

$$\sum F = 0$$
$$F_{thrust} = -F_{drag}$$
$$F_{drag} = -kv$$

Conservation of mass: there is a "continuity" of the mass which is not added or removed. Since the liquid is incompressible, there is no change in density, so $d\rho/dt = 0, \rho = const$. Hence the second equation in Navier-Stokes:

$$\nabla \cdot v = 0$$

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#conservationandmotion: I explain how conservation of momentum and mass apply in the microf
luid situation, and expand it in Navier-Stokes simplification in the next part.
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Calculation

Now I will try to summarize what I found from reading related studies.

We want to solve the Navier-Stokes equations specific to microfluids — low Reynolds number flows. For the simplest model, physicists construct a Purcell Swimmer with Resistive Force Theory. Recently, there are more accurate models like the squirmer which adds more details about how the flagella interact with friction and velocity.

Navier-Stokes equations

for incompressible homogeneous Newtonian fluid:

1. Conservation of mass and momentum can apply to motion in liquid.
2. Navier-Stokes equations describe conservation of momentum and mass in fluids.

$$\rho(\frac{\partial v}{\partial t} + (v \cdot \nabla)v) = -\nabla p + \eta \nabla^2 v$$
$$\nabla \cdot v = 0$$

   Where u is fluid velocity, ρ is fluid density, p is pressure, η is fluid viscosity.

   Incompressible. Mass density is uniform. The only external force assumed is gravity for Newtonian fluids. But later we will ignore gravity term pg because it is much smaller than the viscuous force or it balances out with pressure gradient.

Nondimensionalizing

We can perform the Buckingham Π theorem to make the Navier-Stokes equations dimensionless. I referred to Kirby (n.d.) for the characteristic scale especially for time.

1. The variables are $v_x, v_y, v_z, x, y, z, t, p$.
2. The fundamental dimensions are $[m]$, $[L]$, $[t]$.
3. Characteristic scales: velocity v_o, length L. Characteristic time is the fastest time, $t_o = L/u_o$. Express the original variables in dimensionless forms. Tilde means the quantity is dimensionless.

$$\tilde{v} = \frac{v}{v_o}; \tilde{r} = \frac{r}{L}; \tilde{\nabla} = L\nabla; \tilde{t} = \frac{t}{L/v_o}; \tilde{p} = \frac{p}{\eta v_o / L}$$

4. Substituting the new expressions for the variables into the Navier Stokes equation,

$$\rho((\frac{\partial(\tilde{v}v_o)}{\partial(\tilde{t}t_o)}) + (\tilde{v}v_o) \cdot (\frac{\tilde{\nabla}}{L})(\tilde{v}v_o)) = -(\frac{\tilde{\nabla}}{L})(\frac{\eta v_o \tilde{p}}{L}) + \eta(\frac{\tilde{\nabla}}{L})^2(\tilde{v}v_o)$$

   Group constant terms from dimensionless variables, we get four differentiable terms.

$$\frac{\rho v_o}{t_o}\frac{\partial \tilde{v}}{\partial \tilde{t}} + \frac{\rho v_o}{L}\tilde{v}\cdot\tilde{\nabla}\tilde{v} = -\frac{\eta v_o}{L^2}\tilde{\nabla}\tilde{p} + \eta\frac{v_o}{L^2}\tilde{\nabla}^2\tilde{v}$$

Concentrate the constants to the left hand side by dividing with $\eta v_o/L^2$, and regrouping the terms, we can get dimensionless constants $Re = \frac{\rho v_o L}{\eta}$ and $St = \frac{t_o}{L/v_o}$.

$$\frac{\rho v_o L}{\eta}\frac{L}{v_o t_o}\frac{\partial \tilde{v}}{\partial \tilde{t}} + \frac{\rho v_o L}{\eta}\tilde{v}\cdot\tilde{\nabla}\tilde{v} = -\tilde{\nabla}\tilde{p} + \tilde{\nabla}^2\tilde{v}$$

5. We have a dimensionless Navier-Stokes:

$$\frac{Re}{St}\frac{\partial \tilde{v}}{\partial \tilde{t}} + Re(\tilde{v}\cdot\tilde{\nabla}\tilde{v}) = -\tilde{\nabla}\tilde{p} + \tilde{\nabla}^2\tilde{v}$$

By making the equation dimensionless we found important relationships. The Reynolds number constant lets us talk about the ratio between inertial force and viscous force.

At low Reynolds numbers

The Reynolds number for bacteria swimming in water is very very small, compared to that of humans ($Re = 10^6$). The following estimate comes from Purcell (1977).

Typical bacteria length: $L \approx 1\mu m$

Typical water velocity: $v \approx 30\mu m/s$

Typical water density: $\rho \approx 1g/L$

Typical water viscosity: $\mu \approx 1mPa/s$

$$Re = \frac{\rho v L}{\mu}$$

Reynolds number for bacteria swimming in water: $Re = 3 \times 10^{-5}$.

With this information we can further simplify the Navier-Stokes equations.

We see that as Reynolds number approaches 0 ($\lim_{Re\to 0}$), the two inertial terms on the left hand side approaches 0. We get the Stokes equations.

$$0 = -\nabla p + \eta\nabla^2 u$$
$$\nabla\cdot u = 0$$

Reasoning from the idea that Reynolds number is the ratio between viscous and inertial forces, a low Re implies that the viscous force dominates bodies moving in this flow.

Reasoning from having no inertial term gives a similar consequence: external forces are very negligible. Here, force is linearly proportional to velocity. Similarly, torque is linearly proportional to angular velocity. Because there is no inertia, so without a change in applied force the object in motion will not stay in motion. The equation does not vary with time, so

the bacteria cannot go forward if it is doing reciprical motion (like a person paddling in regular cycles). The bacteria can't stay in motion without putting in energy constantly.

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#DA: I use Buckingham Pi Theorem to nondimensionalize Navier-Stokes equations. Nondimension
alizing helps us extract the Reynolds number constant that simplifies the equations specifi
c to our situation of microswimmers in a viscosity-dominated setting.
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Then we want to solve the above differential equation to describe the how the swimmer propels itself. I will not derive it, but among common solutions the simplest one is the Purcell three-link swimmer, which uses some approximations and boundary conditions to get numerical results.

The Purcell swimmer comes from a classic 1977 lecture about life in low Reynolds numbers. Purcell introduces a body with three linked rods that flaps its two ends like flagella.

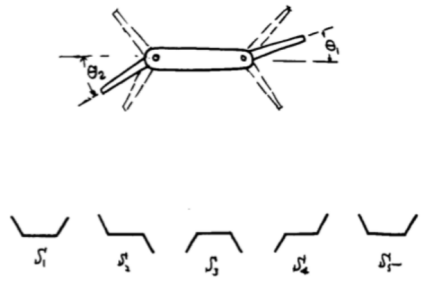

FIG. 14 The Purcell swimmer consists of three rod-like connected segments, which can be tilted with respect to each other. When the two angles θ_1 and θ_2 are varied in the non-reciprocal temporal sequence shown at the bottom, this micro-machine moves forward. From Purcell (1977).

The resistive force theory it describes the relationship between the velocity and the friction force for the Purcell swimmer. Here is a summary of the derivation from Avron et.al. (2008).

It can be expressed as the product of a resistance matrix with the velocity equaling the force.

$$Rv = F$$

Expanding it to get more parameters about the surface for the force:

$$\int_{\Gamma(s)} n \cdot \sigma d\Gamma = -[\frac{2\pi\mu}{\ln(2/\epsilon}](2I - \tau\tau) \cdot v(s) + \mathscr{O}[\ln^{-2}(e/\epsilon)]$$

Where s is arc length along center-line of the swimmer, Γ is the perimeter of the body, ϵ is the fatness— the ratio of diameter to length, τ is the tangential vector, v is the local velocity.

Then we want to numerically implement the solution. The boundary conditions are $\sum F = 0, \sum M = 0$.

Stroking angle, slenderness ratio and hydrodynamic interactions affect the displacement and power efficiency of the swimmer. Experiments found that as paddle's width-to-length ratio increase, displacement per cycle increase.

However, keep in mind that this is a simplified model. The general mechanism is rotating appendages in an order to generate net torque by counter rotation of body and ends.

Conclusion

The actual E Coli is more complicated. It has more rotary motors that are deformable instead of rigid, so it can also change directions by unbundling the flagella and paddle from another side of its body. It experiences Brownian motion which accrlerates the bacteria in circles.

There are also other shapes of swimmers, like larger ones with unevenly distributed density that orient against gravity, or ones with many small flagella instead of few major ones.

As a consequence of our knowledge about bacteria motion, we can design filters accordingly. For one, we can learn from the preference of bacteria swimmer in swimming left to make swirls that draw them into fast filters that then use a vertical reed-sand filter to chemically breakdown compounds and capture by large grains from water. Or, we can reverse-engineer the optimal geometry of micro-swimmer to make robots that capture and decompose large organic compounds in the river in an energy-efficient way.

References

Avron, J. E., & Raz, O. (2008). A geometric theory of swimming: Purcell's swimmer and its symmetrized cousin. New Journal of Physics, 10(6), 063016. https://doi.org/10.1088/1367-2630/10/6/063016

Huber, G., Koehler, S. A., & Yang, J. (2011). Micro-swimmers with hydrodynamic interactions. Mathematical and Computer Modelling, 53(7–8), 1518–1526. https://doi.org/10.1016/j.mcm.2010.04.002

Kirby, B. (n.d.). Nondimensionalization of Navier-Stokes: Reynolds number. Retrieved from http://www.kirbyresearch.com/index.cfm/wrap/textbook/microfluidicsnanofluidicsse137.html

Lauga, E. (2016). Bacterial Hydrodynamics. Annual Review of Fluid Mechanics, 48(1), 105–130. https://doi.org/10.1146/annurev-fluid-122414-034606

Purcell, E. M. (1977). Life at low Reynolds number. American Journal of Physics, 45(1), 3–11. https://doi.org/10.1119/1.10903

Wiezel, O., & Or, Y. (2016). Optimization and small-amplitude analysis of Purcell's three-link microswimmer model. Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences, 472(2192), 20160425. https://doi.org/10.1098/rspa.2016.0425

Sieker, H. (2014). The filter system and how it works. Retrieved from http://www.flussbad-berlin.de/en/-/wie-wird-die-spree-sauber